NFC NS Branch Successor Pilot

v31 — L2 Generated Shell

Stale-Posture Correction: `frontier-blocked` ⇒ `domain-bounded-cert-close`

Generated governance shell

Contents

Governance notice. This is a machine-generated L2 branch-pilot shell. It is *not* a supersession of the canonical NFC Navier-Stokes Branch Book. The canonical NS branch book remains the authoritative source. Full supersession requires L5 gate passage.

NS Branch Successor Pilot

Primary governance function: stale-posture correction. The NS governance repo has been computing `frontier-blocked` (from the stale inference rule `front:NS.main == 0` ⇒ `frontier-blocked`). The canonical NS branch book contains `prop:NS-status [C]`, `thm:NS71 [C]`, and `cor:NS-both-give-endpoint [C]` establishing **domain-bounded conditional CERT-CLOSE**. The NS preamble ("frontier-blocked") is stale relative to the later sections of the same canonical file. This pilot corrects the posture and records the correction as a formal governance finding.

Stale-posture correction record

Stale declared posture: `frontier-blocked`

Corrected computed posture: `domain-bounded-cert-close`

Basis: `thm:NS.7.1 [C]` + `thm:NS.6.1 [C]` + `front:NS.main == 0`

Source: `prop:NS-status [C]` in canonical NS Branch Book

Domain-bounded closure hypotheses

Phase-A, $K_0 = 7$, PASS, LS-2. Removing any hypothesis reopens the domain-bounded closure claim.

Active frontier

`front:NS.main`: Stage-3 global unconditional closure (Navier-Stokes Clay Millennium Prize Problem). This is an *external* frontier outside the current NFC toolkit. No admissible discharge mode exists within the current architecture.

SB2 and Ren.5 status

`ob:NS.SB2`: conditionally discharged [C] via SCT.6b (`thm:NS.SB2-discharged`).

`thm:NS.Ren5`: conditionally discharged [C] via sub-lemma chain.

Table 1: NS successor-pilot claim catalog

ID	**Kind**	**D.**	**C.**	**Scope**
thm:NS.stage0	theorem	U	U	unconditional
thm:NS.stage1	theorem	U	U	unconditional
thm:NS.stage2a	theorem	U	U	unconditional
thm:NS.stage2b	theorem	C	C	conditional-PASS
thm:NS.leray	theorem	U	U	unconditional
thm:NS.window-uniformity	theorem	C	C	conditional-Phase-A
thm:NS.6.1	theorem	C	C	branch-local
thm:NS.Ren5	theorem	C	C	conditional-Phase-A
thm:NS.SB2-discharged	theorem	C	C	conditional-SCT6b
ob:NS.SB2	obligation	C	C	branch-local
ob:NS.bridge-stage2	obligation	O	O	branch-local
thm:NS.7.1	theorem	C	C	domain-bounded
cor:NS.domain-bounded-cert-close	corollary	C	C	domain-bounded
prop:NS.status	proposition	C	C	domain-bounded
front:NS.main	frontier	O	O	branch-local
status:NS	status_note	domain-bounded-cert-close	domain-bounded-cert-close	branch-wide

Generated claim shells

Theorem: NS Stage-0 Thermodynamic Floor Density

ID. `thm:NS.stage0` **Decl.** U **Comp.** U

Scope. regime=NS-source-descended; domain=NS-branch; certificate_scope=unconditional; promotion_ceiling=U

The thermodynamic floor density f_inf > 0 exists unconditionally: under the NS active architecture declarations (N-add, T-add, BV, non-degeneracy), the Van Hove exhaustion window family has a positive limiting floor density. This is the first unconditional structural result of the NS branch and holds without any Phase-A or K_0 hypothesis.

Theorem: NS Stage-1 Boundary Flux Stabilization

ID. `thm:NS.stage1` **Decl.** U **Comp.** U

Scope. regime=NS-source-descended; domain=NS-branch; certificate_scope=unconditional; promotion_ceiling=U

The boundary flux phi_inf >= 0 stabilizes unconditionally: the source-descended boundary flux of the NS observable family achieves a non-negative limit along the Van Hove exhaustion. This follows from the floor density (Stage-0) and the declared monotonicity of the transport regime.

Dependencies. thm:NS.stage0

Theorem: NS Stage-2a Discrete Balance Identity

ID. `thm:NS.stage2a` **Decl.** U **Comp.** U

Scope. regime=NS-source-descended; domain=NS-branch; certificate_scope=unconditional; promotion_ceiling=U

The discrete balance identity holds unconditionally for the source-descended NS obstruction: for every test element f in the observable family, sum_k <D_k, f> = <D_inf, f>. The identity is the canonical decomposition of the obstruction quantity into its transport and defect components. No continuum or PASS hypothesis is needed.

Dependencies. thm:NS.stage1

Theorem: NS Stage-2b Nonlinear Term Identification

ID. `thm:NS.stage2b` **Decl.** C **Comp.** C

Scope. regime=NS-PASS-screened; domain=NS-branch; certificate_scope=conditional-PASS; promotion_ceiling=C; hyp=PASS-screening

The nonlinear term (u dot grad)u + grad p is identified from the discrete balance identity in the continuum limit, conditional on PASS screening: the transverse component of the nonlinear term vanishes under PASS, yielding the canonical Navier-Stokes nonlinear form. This identification requires the PASS condition from Book II and is therefore conditional [C].

Hypotheses. PASS-screening

Dependencies. thm:NS.stage2a

Theorem: NS Leray Weak Existence

ID. `thm:NS.leray` **Decl.** U **Comp.** U

Scope. regime=NS-source-descended; domain=NS-branch; certificate_scope=unconditional; promotion_ceiling=U

For every initial datum u_0 in L^2(R^3) with div u_0 = 0, there exists a global Leray weak solution u in L^inf_t L^2_x intersect L^2_t H^1_x satisfying the energy inequality. The proof proceeds via Galerkin approximation on the NS observable family, using the discrete balance identity (Stage-2a), Aubin-Lions compactness, and PASS screening for incompressibility. This result is unconditional: no Phase-A or K_0 hypothesis is needed. It is the canonical base for all subsequent NS closure arguments.

Dependencies. thm:NS.stage2a

Theorem: NS Window-Uniformity and Eventual Periodicity (Q2a Resolved)

ID. `thm:NS.window-uniformity` **Decl.** C **Comp.** C

Scope. regime=NS-Phase-A-K0-7; domain=NS-branch; certificate_scope=conditional-Phase-A; promotion_ceiling=C; hyp=Phase-A,K0-7,kappa-bound

Under Phase-A, K_0=7, and the kappa-bound (kappa <= 1 - 8/343), the NS active architecture window configurations eventually become periodic: there exists N_0 such that for all n >= N_0, the window configuration W_n is determined by a finite state machine with periodic transitions. The uniform bounds c_1 >= 1/7, c_2 >= 1/49, eta <= 6/7, kappa <= 1 - 8/343 hold. This is the Q2a resolution: the NS active architecture declaration implies deterministic finite-state transition of window configurations, giving periodicity (not merely recurrence). Status: [C] conditional on Phase-A/K_0=7.

Hypotheses. Phase-A, K0-7, kappa-bound

Dependencies. thm:NS.stage2a, thm:NS.6.1

Theorem: Active Missing Bridge Theorem

ID. `thm:NS.6.1` **Decl.** C **Comp.** C

Scope. regime=declared-NS-target; domain=NS-branch; certificate_scope=branch-local; promotion_ceiling=C

Branch-local conditional bridge theorem placeholder for NS.

Dependencies. thm:VI.8.1, front:NS.main

Discharge edges. ob:NS.bridge-stage2 (partial)

Proof stub. Proof placeholder.

Theorem: Late Alignment-to-Dissipation Lemma (Ren.5)

ID. `thm:NS.Ren5` **Decl.** C **Comp.** C

Scope. regime=NS-Phase-A-K0-7; domain=NS-branch; certificate_scope=conditional-Phase-A; promotion_ceiling=C; hyp=Phase-A,K0-7,PASS

The Late Alignment-to-Dissipation Lemma (Ren.5): under Phase-A, K_0=7, and PASS, any infinite late counterexample to the alignment-to-dissipation property is eliminated by three-branch case analysis (sub-lemma chain NS.Ren.5a through NS.Ren.5e: Adaptation Dominance, Renewal-vs-Adaptation Trichotomy, Averaging Attenuation, Active Admissibility). All three branches being eliminated, Ren.5 holds inside the active architecture. The sub-lemma chain is conditional [C]. This discharges the proof-program frontier of the NS branch.

Hypotheses. Phase-A, K0-7, PASS

Dependencies. thm:NS.window-uniformity, thm:NS.6.1

Theorem: SB2 Conditionally Discharged via SCT.6b

ID. `thm:NS.SB2-discharged` **Decl.** C **Comp.** C

Scope. regime=NS-Phase-A-K0-7; domain=NS-branch; certificate_scope=conditional-SCT6b; promotion_ceiling=C; hyp=Phase-A,K0-7,PASS,SCT.6b-firing-conditions

SB2 (Positive Reserve on the Safe Tail) is conditionally discharged via the SCT.6b coercive-transport mechanism: once a tail window satisfies the three SCT.6b firing conditions (coercive structure, transport compatibility, subcriticality check), SB2 is discharged from the assembled proof record via the shared coercive-transport doctrine (SCT.1-SCT.6b). The full discharge requires Ren.5 (thm:NS.Ren5) as the upstream input. Discharge mode: full, at the conditional scope of Phase-A/K_0=7/PASS/SCT.6b.

Hypotheses. Phase-A, K0-7, PASS, SCT.6b-firing-conditions

Dependencies. thm:NS.Ren5

Discharge edges. ob:NS.SB2 (full)

Obligation: Positive Reserve on the Safe Tail

ID. `ob:NS.SB2` **Decl.** C **Comp.** C

Scope. regime=declared-NS-target; domain=NS-branch; certificate_scope=branch-local; promotion_ceiling=C

Positive Reserve on the Safe Tail (SB2): CONDITIONALLY DISCHARGED via SCT.6b (thm:NS.SB2-discharged). The discharge is conditional on Phase-A/K_0=7/PASS and the SCT.6b firing conditions. The obligation is closed at this scope. Unconditional force for SB2 would require O-IDcont-style force upgrade (external to current architecture).

Obligation: NS Bridge Stack Stage 2 Obligation

ID. `ob:NS.bridge-stage2` **Decl.** O **Comp.** O

Scope. regime=declared-NS-target; domain=NS-branch; certificate_scope=branch-local

Intermediate NS bridge-stage obligation used to test explicit discharge edges.

Dependencies. thm:VII.4.3

Theorem: Continuation-Criterion-to-Endpoint Theorem (NS.7.1)

ID. `thm:NS.7.1` **Decl.** C **Comp.** C
Scope. regime=NS-Phase-A-K0-7-PASS-LS2; domain=NS-branch; certificate_scope=domain-bounded; promotion_ceiling=C; hyp=Phase-A,K0-7,PASS,LS-2,NS.6.1-discharged,Book-VI-H1x-interface

NS.7.1 (Continuation-Criterion-to-Endpoint Theorem): if the active NS persistence/extendibility continuation criterion holds in the promoted target scope (established by NS.6.1), then the NS endpoint target — the Leray weak solution u in L^inf_t H^1_x on the declared interval — follows. The proof combines: (1) Gronwall control from the multiplicative contraction D_{n+1} <= kappa^{n-n_0} D_{n_0} -> 0 (giving summable obstruction); (2) Book VI H^1_x-norm legitimacy (licensing the smooth-notation statement). Domain-bounded CERT-CLOSE is achieved under Phase-A/K_0=7/PASS/LS-2. NS.7.1 is structurally separate from NS.6.1 by canonical discipline: continuation and regularity are independently auditable.

Hypotheses. Phase-A, K0-7, PASS, LS-2, NS.6.1-discharged, Book-VI-H1x-interface

Dependencies. thm:NS.6.1, thm:NS.window-uniformity, thm:NS.SB2-discharged

Corollary: NS Domain-Bounded CERT-CLOSE

ID. `cor:NS.domain-bounded-cert-close` **Decl.** C **Comp.** C
Scope. regime=NS-Phase-A-K0-7-PASS-LS2; domain=NS-branch; certificate_scope=domain-bounded; promotion_ceiling=C; hyp=Phase-A,K0-7,PASS,LS-2

The NS endpoint target (Leray weak solution u in L^inf_t H^1_x) follows from the declared source-descended obstruction-decay law, conditionally. The NS branch achieves domain-bounded CERT-CLOSE on the declared interval under Phase-A/K_0=7/PASS/LS-2. Proof: Corollary of NS.6.1 (conditional) composed with NS.7.1 (conditional); both are [C]; their composition is [C]. Unconditional global regularity (Stage-3 Clay Prize) remains the external frontier.

Hypotheses. Phase-A, K0-7, PASS, LS-2

Dependencies. thm:NS.6.1, thm:NS.7.1

Proposition: NS Branch Status (Domain-Bounded Conditional CERT-CLOSE)

ID. `prop:NS.status` **Decl.** C **Comp.** C
Scope. regime=NS-Phase-A-K0-7-PASS-LS2; domain=NS-branch; certificate_scope=domain-bounded; promotion_ceiling=C; hyp=Phase-A,K0-7,PASS,LS-2

NS Branch Status (Updated): The NS branch is a lawful descendant of the canonical spine and has reached domain-bounded conditional CERT-CLOSE under Phase-A/K_0=7/PASS/LS-2 hypotheses. NS.6.1 is conditionally discharged (window-uniformity + K_0=7 + Phase-A + PASS + kappa-bound). Ren.5 is conditionally discharged (sub-lemma chain NS.Ren.5a-5e). SB2 is conditionally discharged (via SCT.6b). NS.7.1 is conditionally discharged (Gronwall + Book VI H^1_x interface). Domain-bounded CERT-CLOSE is achieved. The NS Frontier Theorem (thm:NS-frontier [U]) describes the formal conditional frontier; under the stated hypotheses the branch reaches closure. Unconditional global regularity (NS Stage-3 Clay Prize) remains the external frontier. Note: the NS preamble ("frontier-blocked") predates this discharge chain and is stale — the corrected posture is domain-bounded conditional CERT-CLOSE.

Hypotheses. Phase-A, K0-7, PASS, LS-2

Dependencies. cor:NS.domain-bounded-cert-close, thm:NS.SB2-discharged, thm:NS.Ren5

Frontier: NS Main Frontier

ID. `front:NS.main` **Decl.** O **Comp.** O
Scope. regime=declared-NS-target; domain=NS-branch; certificate_scope=branch-local; promotion_ceiling=O

Stage-3 global unconditional closure: prove unconditional global regularity for all smooth initial data for the 3D Navier-Stokes equations (the Clay Millennium Prize Problem). The domain-bounded conditional closure has been achieved under Phase-A/K_0=7/PASS/LS-2 (thm:NS.7.1). This frontier records the residual gap between domain-bounded conditional CERT-CLOSE and unconditional global CERT-CLOSE. It is an external frontier: its resolution requires tools outside the current NFC active architecture. No admissible discharge mode exists within the current canonical toolkit.

Status note: NS Branch Posture Record

ID. `status:NS` **Decl.** domain-bounded-cert-close **Comp.** domain-bounded-cert-close
Scope. regime=ns-branch-governance; domain=NS-branch; certificate_scope=branch-wide; promotion_ceiling=fron
blocked

NS branch posture: domain-bounded-cert-close (conditionally). Correction note: the NS preamble says "frontier-blocked" but this is stale relative to the later sections of the canonical NS branch book, which establish domain-bounded conditional CERT-CLOSE via prop:NS.status [C]. The governance pilot records this stale-posture correction as a formal governance finding.

Active frontier: front:NS.main (Stage-3 global unconditional closure = Clay Prize). This is an external frontier outside the current NFC toolkit.

The domain-bounded closure is conditional on: Phase-A, K_0=7, PASS, LS-2. Removing any of these hypotheses reopens the domain-bounded closure claim.

Dependencies. prop:NS.status